

Project Documentation

PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Project Overview

PocketSmart AI is a Generative AI-powered budget and recommendation assistant.

It helps users plan expenses and receive personalized recommendations for:

- Home Interior
- Party Planning
- Jewelry Selection

The system combines FastAPI with Gemini 1.5 Flash Pro and a web-based frontend.

2. Key Features

Authentication

- User registration
- Login
- Logout
- JWT authentication
- Session management

Home Interior Planner

Provides budget-based recommendations for home decoration and interior products.

Party Planner

Helps users plan events according to budget and guest count.

Jewelry Planner

Provides jewelry recommendations based on budget, occasion, style and optional outfit images.

Recommendation History

Allows users to view previous recommendation queries and results.

3. Technology Stack

Frontend:

HTML + CSS + JavaScript + Jinja2

Backend:

Python + FastAPI

AI:

Gemini 1.5 Flash Pro

Server:

Uvicorn

Authentication:

JWT

External Platforms:

Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO

4. User Workflow

Register / Login

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Dashboard

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Select Planner

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Enter Requirements

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Submit Request

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FastAPI Backend

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Gemini AI

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Generate Recommendations

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View Results

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Save / Review History

5. Planner Documentation

Home Interior Planner

The user provides budget, room information and product requirements.

The system processes the information and provides suitable interior recommendations.

Party Planner

The user provides budget, guest count, event type and other requirements.

The system provides recommendations for food, venues and decorations.

Jewelry Planner

The user provides budget, occasion and style preferences.

An outfit image can optionally be uploaded for additional visual context.

6. API Documentation

Authentication APIs

Endpoint	Purpose
<code>/register</code>	Register a new user
<code>/login</code>	Authenticate user
<code>/logout</code>	End user session
<code>/token</code>	Generate authentication token

Planner APIs

Endpoint	Purpose
<code>/generate-home</code>	Home recommendations
<code>/generate-party</code>	Party recommendations
<code>/generate-jewelry</code>	Jewelry recommendations

Session APIs

Endpoint	Purpose
<code>/session-info</code>	Current session information
<code>/session-data</code>	User-specific session data
<code>/history</code>	Previous recommendations
<code>/recommendations-details</code>	Detailed recommendation results

7. Benefits

PocketSmart AI aims to:

- Save users' time.

- Simplify budget planning.
 - Provide personalized recommendations.
 - Reduce manual product searching.
 - Combine multiple planning categories in one platform.
 - Support image-based recommendation for jewelry.
 - Improve decision-making through AI assistance.
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8. Future Scope

Potential future improvements include:

- More planning categories.
 - More external platforms.
 - Real-time product availability.
 - Real-time price comparison.
 - Advanced personalization.
 - Improved recommendation ranking.
 - Mobile application support.
 - Additional AI capabilities.
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9. Conclusion

PocketSmart AI demonstrates how Generative AI can be integrated with modern web technologies to create a practical budget-aware recommendation platform.

By combining FastAPI, Gemini 1.5 Flash Pro and an interactive frontend, the system provides personalized assistance for everyday lifestyle planning.